

CSDA5110
Homework 6
Fall 1 2024

- 1) Creatinine clearance (Y) is an important measure of kidney function, but is difficult to obtain in a clinical office setting because it requires 24-hour urine collection. To determine whether this measure can be predicted from some data that are easily available, a kidney specialist obtained the data from 33 male subjects. The predictor variables are serum creatinine concentration (X_1), age (X_2), and weight (X_3).
- Obtain the scatter plot matrix and the correlation matrix. What do the scatter plots suggest about the nature of the functional relationship between the response variable Y and each predictor variable?
 - Fit the regression model to the data. State the estimated regression function.
 - Does it appear that all predictor variables should be retained?
 - Using first-order and second-order terms for each of the three predictor variables in the pool of potential X variables (including cross products of the first-order terms), find the **three best** subset regression models according to the C_p criterion.
 - Is there much difference in C_p for the three best subset models?
 - Using the same pool of potential X variables as in part(d), find the best subset of variables according to forward stepwise regression.
 - Using the same pool of potential X variables as in part(d), find the best subset of variables according to backward elimination regression.
 - How does the best subset according to these three methods compare with respect to C_p and adjusted R^2 ?
 - Obtain the studentized delete residuals and identify any outlying Y observations. Use the Bonferroni outlier test procedure with $\alpha = .01$.
 - Obtain the DFFITS, DFBETAS, and Cook's distance values for each case to assess its influence. What do you conclude?
 - Calculate Cook's distance D for each case and prepare an index plot. Are any cases influential according to this measure'?